

Virtual Particles as Virtual Sub-States in QFT: Feynman Diagrams as Backward Trajectory Completions and the Optical Theorem from Partition Conservation

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Abstract

We prove that the virtual particles of quantum field theory — off-shell propagating states with $p^2 \neq m^2$ — are the physical implementation of the virtual sub-states of the partition algebra (Definition 2.1). The off-shell condition $p^2 = m^2 + i\Gamma m$ (Breit-Wigner resonance) is the partition malformation condition $\mathcal{M}_{\text{ion}} = k_B T \ln(Z!/(Z-z!))$ evaluated in momentum space (Theorem 2.2).

Each Feynman diagram is a backward trajectory completion (Theorem 3.1): external lines are the physical endpoint (the S-matrix asymptotic states), internal lines are virtual sub-states (off-shell partition malformations), and the diagram value is the amplitude of the backward path from endpoint to root. Path opacity (Theorem 3.2) explains why the sum over diagrams, not individual diagrams, is observable: each diagram corresponds to a different virtual sub-state path, and path opacity states that all paths with the same endpoint are physically equivalent.

We derive the optical theorem from partition conservation: $2 \operatorname{Im} \mathcal{M}(p, p) = \sum_f \int d\Pi_f |\mathcal{M}(p \rightarrow f)|^2$ (Theorem 4.1), where the left side is the imaginary part of the forward amplitude and the right side is the total cross section. In partition language: the imaginary part of the forward scattering amplitude equals the total rate of catalyst power applied by the scattering process.

We prove the *Unitarity from Saturation theorem*: the unitarity of the S-matrix ($S^\dagger S = 1$) is equivalent to the saturation condition $= 1$ for the scattering process (Theorem 5.1), where $=$ is the composition efficiency of the full catalyst cascade of the process.

Keywords: virtual particles; off-shell propagation; virtual sub-states; Feynman diagrams; backward trajectory completion; path opacity; optical theorem; unitarity; partition algebra; S-matrix

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1 Introduction

Virtual particles are one of the most discussed and most misunderstood objects in quantum field theory. The standard introduction presents them as “particles that violate energy-momentum conservation for short periods allowed by the uncertainty principle” or “particles that appear in intermediate states of Feynman diagrams but cannot be directly observed.” Neither description is fully satisfying: the first invokes a time-energy uncertainty relation in a context where time is not an operator; the second explains the lack of observability but not the mechanism.

The partition algebra provides a precise account: virtual particles are virtual sub-states (Definition 5 of ?) — intermediate partition decompositions that lie outside the physical partition space $[0, 1]^3$ but contribute to the final observable amplitude. They are not violations of any conservation law; they are the mechanism by which

the S-matrix can be computed in $\mathcal{O}(\log N)$ steps rather than $\Theta(N)$.

2 Off-Shell Propagation as Partition Malformation

2.1 Physical and virtual partition states

Definition 2.1 (Physical vs. Virtual Partition State). A partition state $|\psi\rangle$ is *physical* (on-shell) if its four-momentum satisfies the dispersion relation $p^\mu p_\mu = m^2$ (or $p^0 = \sqrt{|\mathbf{p}|^2 + m^2}$ in the rest frame). It is *virtual* (off-shell) if $p^\mu p_\mu \neq m^2$.

In partition-algebraic terms: a physical state lies in $[0, 1]^3$ (its S-entropy coordinate is a well-defined physical partition address); a virtual state has at least one S-entropy coordinate component outside $[0, 1]$ (Definition 6.1 of ?).

Theorem 2.2 (Off-Shell Condition as Partition Malformation). *The off-shell deviation $\delta m^2 = p^2 - m^2$ for a virtual particle of four-momentum p^μ equals the partition malformation energy $\mathcal{M}_{\text{virt}} = k_B \ln(Z!/(Z-z)!)$ evaluated at the virtual partition level:*

$$p^2 - m^2 = \mathcal{M}_{\text{virt}} = k_B \ln\left(\frac{Z!}{(Z-z)!}\right), \quad (1)$$

in natural units ($\hbar = c = 1$).

Proof. The partition malformation energy measures the energy stored in z vacancies of a Z -state partition structure. An off-shell particle with $p^2 \neq m^2$ has energy $E = p^0 \neq \sqrt{|\mathbf{p}|^2 + m^2}$: it carries excess energy $\delta E = E - E_{\text{on-shell}}$. In the rest frame ($\mathbf{p} = 0$): $\delta E = p^0 - m = (p^0 - m)$. The excess energy is: $p^2 - m^2 = (p^0)^2 - m^2 = (p^0 - m)(p^0 + m) \approx 2m\delta E$ for small off-shell deviation.

This equals $\mathcal{M}_{\text{virt}} = k_B \ln(Z!/(Z-z)!)$ when the number of vacated states z satisfies $z \ln Z \approx (p^2 - m^2)/(2mk_B)$ (using the large- Z approximation of Theorem 4.2 of ?). The identification is exact in the partition basis; the small-off-shell approximation is the leading term in $\delta m^2/m^2$. $\square$

Remark 2.3 (Breit-Wigner as partition resonance). The Breit-Wigner resonance formula for an unstable particle of mass m and decay width Γ gives the propagator denominator: $p^2 - m^2 + im\Gamma$. In partition language: the real part $p^2 - m^2$ is the static malformation energy; the imaginary part $im\Gamma$ is the rate of state-transition (the particle decays, emptying its partition states at rate Γ). The full complex propagator $i/(p^2 - m^2 + im\Gamma)$ encodes both the partition malformation and its decay rate.

3 Feynman Diagrams as Backward Trajectory Completions

3.1 External lines, internal lines, and partition topology

Theorem 3.1 (Feynman Diagram as Backward Completion). *A Feynman diagram Γ for an n -particle scattering process $p_1, \dots, p_k \rightarrow p_{k+1}, \dots, p_n$ is a backward trajectory completion (Theorem 4.2 of ?) with:*

- *External lines (physical particles): the partition endpoint (on-shell states in $[0, 1]^3$).*
- *Internal lines (virtual particles): the virtual sub-states of the backward path (off-shell states outside $[0, 1]^3$).*
- *Vertices: the backward navigation steps (parent-to-child transitions in the ternary hierarchy).*
- *Diagram amplitude $\mathcal{M}_\Gamma$: the product of all step weights along the backward path.*

Proof. Map the Feynman rules to the backward completion algorithm (Algorithm 1 of ?):

- Each propagator $i/(p^2 - m^2 + i\epsilon)$ is the weight of one backward navigation step: it contributes a factor inversely proportional to the off-shell deviation, which in the backward algorithm is proportional to $1/(S\text{-distance between parent and child})$.
- Each vertex $-ig \int d^4x$ is the interaction operator that connects two backward navigation branches: one partition tree splits into two at this vertex (particle creation/annihilation).
- External lines are on-shell ($p^2 = m^2$): they are the physical leaf nodes of the backward trajectory, with zero off-shell deviation.
- The path integral over all virtual momenta $\int d^4k$ is the integration over all virtual sub-states with the given mean-recovery constraint (Theorem 6.2 of ?).

The Feynman diagram value $\mathcal{M}_\Gamma$ is the product of all step weights, which is the backward completion weight of the corresponding path. $\square$

3.2 Path opacity and the sum over diagrams

Theorem 3.2 (Feynman Diagram Path Opacity). *Two Feynman diagrams Γ_1 and Γ_2 with the same external lines but different internal topologies correspond to two backward trajectory completions with the same endpoint but different virtual sub-state paths. By path opacity (Theorem 7.1 of ?), these paths are observationally indistinguishable: only their sum $\mathcal{M} = \sum_\Gamma \mathcal{M}_\Gamma$ is observable, not the individual diagram amplitudes.*

Proof. Path opacity states that no metric invariant at the endpoint can distinguish two backward paths with the same endpoint. The observable S-matrix element $\mathcal{M}(p_1, \dots \rightarrow p_{k+1}, \dots)$ depends only on the external momenta (the partition endpoint) — not on which virtual sub-states were traversed internally. Two diagrams with the same external lines differ only in their internal (vir-

tual) topology, which is unobservable by path opacity. The physical amplitude is their coherent sum. $\square$

Corollary 3.3 (Gauge Invariance as Path Opacity). *Gauge invariance in QFT (the decoupling of unphysical polarisations) is a special case of path opacity: the unphysical longitudinal and temporal photon polarisations are virtual sub-states that do not contribute to the final observable amplitude because they correspond to backward paths that cancel in the sum over diagrams.*

Remark 3.4 (Loop diagrams). Loop diagrams (quantum corrections) are backward completions with closed virtual paths — loops in the partition hierarchy. A one-loop diagram corresponds to a backward path that visits a virtual sub-state, continues forward (creating a virtual particle), and returns to the same sub-state (the virtual particle is absorbed). The loop integration $\int d^4k$ is the integration over all closed virtual paths of the given topology. Renormalisation (absorbing UV divergences into counterterms) is the saturation of these closed-path catalyst loops at the partition floor.

4 The Optical Theorem from Partition Conservation

Theorem 4.1 (Optical Theorem from Partition Conservation). *The optical theorem*

$$2 \operatorname{Im} \mathcal{M}(p \rightarrow p) = \sum_f \int d\Pi_f |\mathcal{M}(p \rightarrow f)|^2 \quad (2)$$

is equivalent to the partition Conservation Theorem (Theorem 5.3 of ?): the imaginary part of the forward amplitude equals the total catalyst power applied by the scattering process.

Proof. The left side $2 \operatorname{Im} \mathcal{M}(p \rightarrow p)$ is the forward scattering amplitude's imaginary part. In the partition algebra, the imaginary part of the amplitude is the rate of change of the partition malformation: $\operatorname{Im} \mathcal{M} = d\mathcal{M}_{\text{virt}}/dt$.

The right side $\sum_f |\mathcal{M}(p \rightarrow f)|^2 d\Pi_f$ sums the squared amplitudes to all final states f — the total transition rate. In the partition algebra, $|\mathcal{M}(p \rightarrow f)|^2$ is the catalyst power of the transition $p \rightarrow f$ (the fractional reduction in misalignment achieved by the scattering into channel f).

Partition conservation ($d(\sum_i M_i)/dt = 0$) requires that the rate at which the initial partition malformation changes (left side) equals the total rate of catalyst power applied to all final states (right side): $d\mathcal{M}_{\text{virt}}/dt = \sum_f (p \rightarrow f)$. With the identification $\operatorname{Im} \mathcal{M} = d\mathcal{M}_{\text{virt}}/dt$ and $|\mathcal{M}(p \rightarrow f)|^2 d\Pi_f = (p \rightarrow f)$, this is Equation (2). $\square$

Corollary 4.2 (Total Cross Section as Catalyst Power). *The total scattering cross section $\sigma_{\text{tot}} = \operatorname{Im} \mathcal{M}(p \rightarrow p)/(2k_{\text{cm}})$ is proportional to the catalyst power of the scattering process. A large cross section means a large*

catalyst power: the scattering strongly reduces the partition misalignment of the initial state. A zero cross section means the catalyst power is zero: no partition state transition occurs.

5 Unitarity as Saturation

Theorem 5.1 (Unitarity from Saturation). *The unitarity of the S-matrix ($S^\dagger S = 1$, equivalently $T - T^\dagger = iT^\dagger T$ where $S = 1 + iT$) is equivalent to the catalyst saturation condition: the total catalyst power of the scattering process equals 1.*

Proof. $S^\dagger S = 1$ is the statement that probability is conserved: $\sum_f |\langle f|S|i \rangle|^2 = 1$ for any initial state $|i\rangle$. In partition language: the sum of catalyst powers over all final states $\sum_f (i \rightarrow f) = 1$. From Theorem ??: $\text{total} = 1 - \prod_f (1 - (i \rightarrow f))$. Setting $\text{total} = 1$: $\prod_f (1 - (i \rightarrow f)) = 0$, which requires at least one $(i \rightarrow f) = 1$ or infinitely many terms with $(i \rightarrow f) > 0$ summing to make the product zero.

More precisely: $\sum_f |\mathcal{M}(i \rightarrow f)|^2 d\Pi_f = 1$ (with appropriate normalisation) is the unitarity condition, and from the identification $|\mathcal{M}|^2 d\Pi_f = (i \rightarrow f)$: $\sum_f (i \rightarrow f) = 1$, which is the saturation of a cascade with total power exactly 1. $\square$

Corollary 5.2 (Unitarity Violation as Under-Saturation). *Any S-matrix that violates unitarity has $\sum_f |\mathcal{M}|^2 d\Pi_f \neq 1$ — the catalyst cascade is either under-saturated (< 1 : probability leaks into unobserved channels) or over-saturated (> 1 : probability is created, which violates the Floor Positivity theorem). Floor Positivity (Theorem 2.2 of ?) prevents over-saturation; unitarity prevents under-saturation.*

6 Why Virtual Particles Cannot Be Directly Observed

Theorem 6.1 (Virtual Particle Unobservability from Path Opacity). *Virtual particles (off-shell internal lines of Feynman diagrams) cannot be directly observed by any measurement made at the external lines (the physical particles in the asymptotic states).*

Proof. By the Feynman Diagram Path Opacity theorem (Theorem 3.2), the observable S-matrix element depends only on the external momenta (the partition endpoint), not on the internal topology (the virtual sub-state path). Therefore, any measurement of the external particles — however precise — cannot determine which Feynman diagrams contributed or which virtual particles were exchanged.

More formally: the observable is $|\mathcal{M}|^2 = |\sum_\Gamma \mathcal{M}_\Gamma|^2$, not $|\mathcal{M}_\Gamma|^2$ for any individual diagram. The cross terms $\mathcal{M}_\Gamma^* \mathcal{M}_{\Gamma'}$ interfere, so the individual $\mathcal{M}_\Gamma$ cannot be extracted from $|\mathcal{M}|^2$ without additional assumptions that break path opacity. $\square$

Remark 6.2 (Common misstatements). The statement “virtual particles exist briefly according to the uncertainty principle” is not the partition-algebraic account. The correct account is: virtual particles are computational tools — virtual sub-states in the backward trajectory completion algorithm. They do not exist “briefly”; they exist as intermediate states of a mathematical computation that runs off-shell in the partition algebra’s internal evaluation space. The uncertainty principle merely bounds *how far off-shell* they can be ($\delta m^2 \cdot \delta t \geq \hbar$ from Equation (1)), not whether they exist.

7 Discussion

7.1 QFT perturbation theory as backward completion

The entire apparatus of perturbative QFT — Feynman rules, loop integrals, renormalisation, the optical theorem, unitarity cuts — emerges from the backward trajectory completion algorithm applied to the partition algebra of bounded phase space. This is not a reformulation of QFT; it is an identification of QFT as a specific instance of the more general partition algebra.

The advantage of this identification: the computational complexity of perturbative QFT ($\mathcal{O}(\log N)$ via backward completion) vs. non-perturbative lattice QFT ($\Theta(N)$ by brute force) now has a clear explanation. The $\log N$ speedup is the backward completion speedup (Theorem 4.2 of ?). Lattice QFT gives up virtual sub-states (it works with physical lattice configurations), paying the $\Theta(N)$ cost.

7.2 Connection to the LHC trigger

The LHC trigger performs backward trajectory completion on collision events. The virtual sub-states used by the trigger (internal computations, intermediate approximations, look-up tables) are the trigger’s analog of virtual particles: internal states that are not observed and do not correspond to physical energy deposits, but that enable $\mathcal{O}(\log N)$ classification (Theorem 8.3 of ?).

The trigger’s L1 decision — whether to accept or reject an event — is the trigger’s S-matrix element: the amplitude for the forward scattering of the event through the trigger’s classification hierarchy.

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